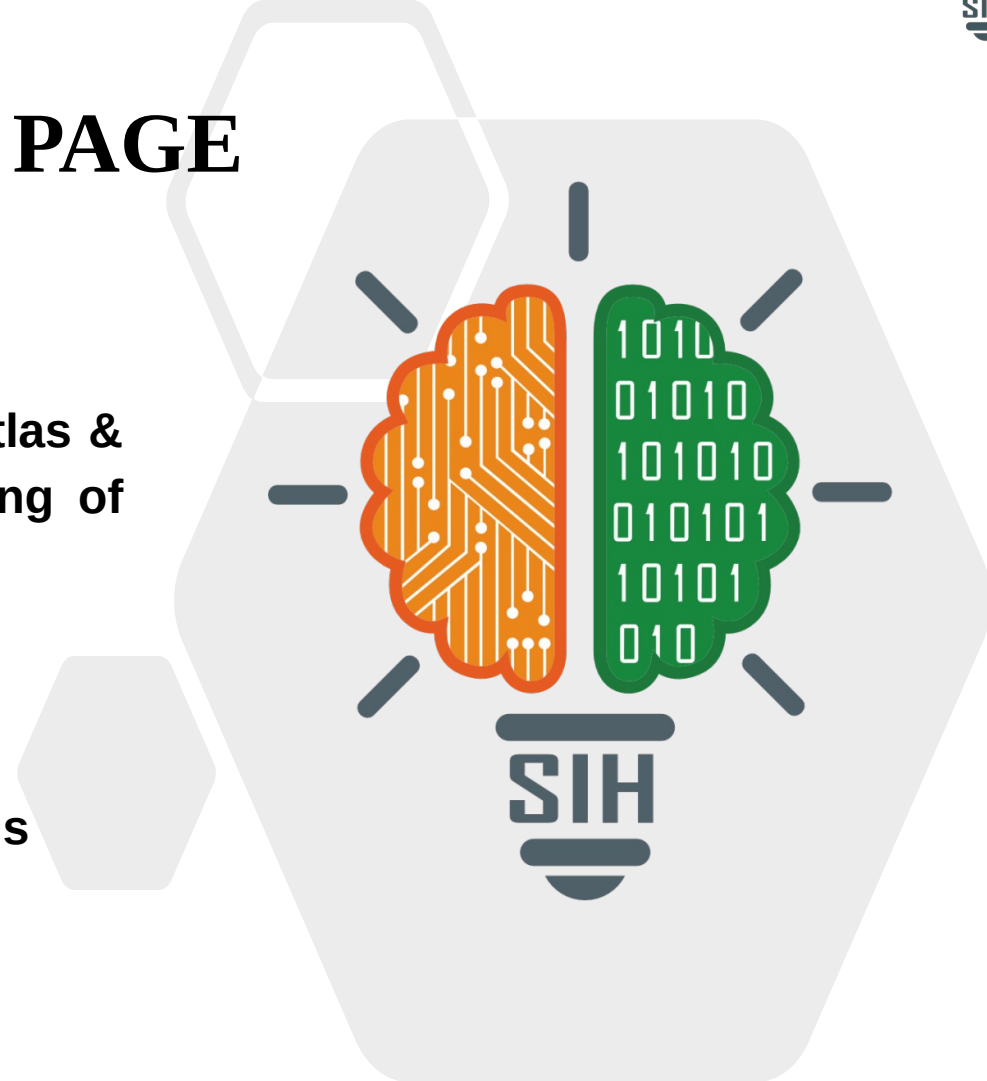




TITLE PAGE

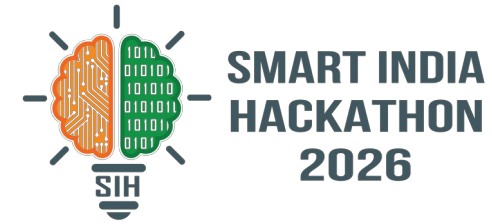
- Problem Statement ID – SIH12508
- Problem Statement Title- AI-Powered FRA Atlas & WebGIS-based DSS for Integrated Monitoring of FRA Implementation
- Theme- Miscellaneous
- PS Category- Software
- Team ID- <Enter your official Team ID>
- Team Name (Registered on portal) - Byte Owls



❖ AI-Powered FRA Atlas & WebGIS Decision Support System

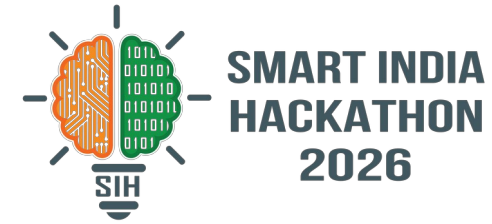
- Live claim mapping, automated conflict detection, and offline-first field verification for low-connectivity forest terrain — turns FRA implementation from a paper backlog into a connected, transparent system for citizen, field officer and district office alike. Phase 1 (Auth, RBAC, Security) is built and verified; Phase 2 (GIS Atlas) is in progress.
- 36.33% of all 51.57 lakh FRA claims filed nationally have been rejected — mostly from missing evidence (39,906 of 46,090 rejections in J&K alone). Our Evidence Completeness Checker and Conflict Engine target this failure point directly, before submission.
- An explainable “Why This Claim?” risk breakdown instead of a black box; a rules-based Scheme Recommendation module covering the full official PS scope; and an honest Trust Center that shows only what is genuinely built — never a fake status indicator.

TECHNICAL APPROACH



- Frontend: React + TypeScript + Vite + Tailwind + Leaflet. Backend: Node.js + Express + TypeScript + Prisma. Database: PostgreSQL + PostGIS (Dockerized). Security: JWT + refresh rotation + bcrypt + RBAC across 4 roles.
- Pipeline: Claim Submitted → Evidence Check → GIS Conflict Detection → Risk Prioritization → Offline Verification → DSS Decision. Built in verified phases — Foundation → GIS Atlas → Claim Engine → Conflict Detection → Risk Engine → Offline Mode → Hardening → Demo Prep — each manually tested before the next begins. Prototype screenshots (FRA Atlas map, RBAC Test Bench, Trust Center) to be added as later phases complete.

FEASIBILITY AND VIABILITY



- Every tool in this prototype runs on a free tier — Leaflet/OpenStreetMap, CARTO, Tesseract.js OCR, Dockerized PostGIS. Zero cost to build; production scaling becomes the deploying government body's call.
- PostGIS spatial query complexity at scale; offline sync conflicts on flaky networks; GPS spoofing risk in field evidence; evaluator distrust of “black-box” AI scoring.
- Risk mitigation: prototype-labeled overlap thresholds tested with unit tests; idempotency keys + version checks so sync never blindly overwrites; GPS spoofing named openly as a known limitation (Phase 2 cross-check planned); and a “Why This Claim?” breakdown that shows every scoring factor plainly.

IMPACT AND BENEFITS

- Target users: tribal/forest-dwelling claimants, field officers, district and state admins under the Forest Rights Act, 2006. Scale: 51.57 lakh FRA claims filed nationally, of which 36.33% are rejected outright — largely from preventable evidence gaps VanSetu is built to close.
- Social: transparency and shorter waits for tribal claimants. Governance: conflict visibility before disputes escalate. Operational: verified field work even with zero connectivity. Scalability: rules-based engines and free-tier stack scale to state-wide claim volumes without added licensing cost.

RESEARCH AND REFERENCES

- Official Problem Statement: SIH12508, Ministry of Tribal Affairs — sih.gov.in. FRA Implementation Data: MoTA claim statistics — 51.57 lakh claims filed, 48.76% approved, 36.33% rejected nationally (as of Oct 2025). Legal Basis: The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 and FRA Rules 2007 (amended 2012). Data Protection Basis: DPDP Act 2023, referenced for claimant data-handling design principles.